# Unit 6, Lesson H3: Justifying Similarity

## Benchmark

MA.912.GR.2.9 Justify the criteria for triangle similarity using the definition of similarity in terms of non-rigid transformations.

## Learning Target

- ☐ I can use the definition of similarity to justify that SSS similarity, SAS similarity, and AA similarity are sufficient to show that two triangles are similar.

## 6.H3.1 Warm-Up: Which One Doesn't Belong?

**Image A**

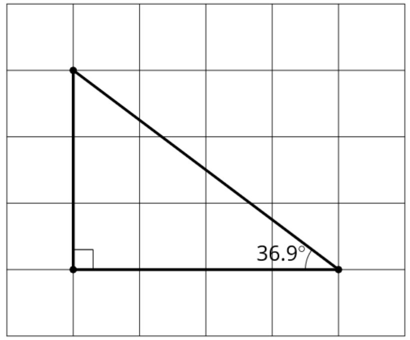



**Image B**

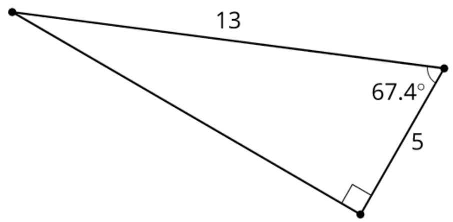



**Image C**

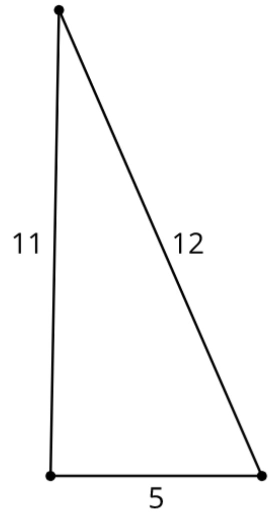
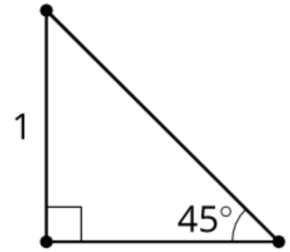


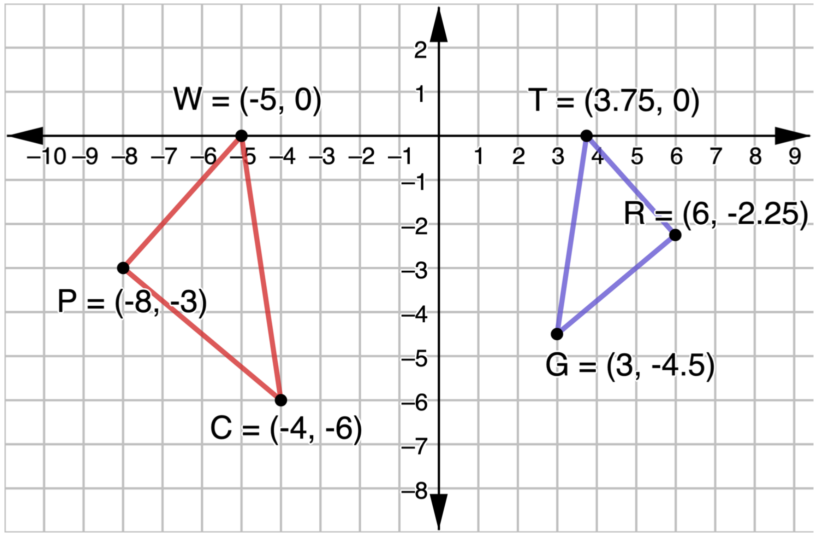

**Image D**




1. Choose an image that does not belong and explain your choice.

## 6.H3.2 Collaboration: Justifying Similarity

1. $\Delta PWC$ and $\Delta RTG$ are shown.




   a. What do you notice about the two triangles?

   b. What transformations can be applied so that $\Delta WPC$ is mapped to $\Delta TRG$?

   c. What can you conclude from the transformations?

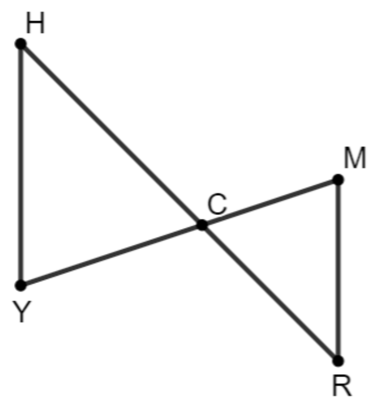
Omario's and Ernest's answers to the question "If $\Delta PWC$ and $\Delta RTG$ are similar by angle-angle similarity, what information would need to be given?" are shown.

| Omario | Ernest |
|---|---|
| $\angle WPC \cong \angle TRG$ and $\angle WCP \cong \angle TGR$ | $\angle PWC \cong \angle RTG$ and $\angle PCW \cong \angle RGT$ |

   d. What do you notice about their answers?

   e. Explain why both students correctly answered the question.

2. $\Delta MRC$ and $\Delta YHC$ are shown.




a. Complete the table.

| Given Information | True or False? $\Delta MRC \sim \Delta YHC$ by AA Similarity |
|---|---|
| $\angle HCY \cong \angle RCM$<br>$\angle H \cong \angle R$ | ○ True<br>○ False |
| $\angle HCY \cong \angle RCM$<br>$\frac{MR}{YH} = \frac{MC}{YC}$ | ○ True<br>○ False |
| $\angle H \cong \angle R$<br>$\angle Y \cong \angle M$ | ○ True<br>○ False |

b. Select all of the transformations in a sequence that could be applied so that $\Delta MRC$ coincides with $\Delta YHC$:

- ☐ rotate $\Delta MRC$ about point $C$
- ☐ rotate $\Delta MRC$ about point $R$
- ☐ reflect $\Delta MRC$ across $\overline{MR}$
- ☐ reflect $\Delta MRC$ across $\overline{YH}$
- ☐ translate $\Delta MRC$ left and up
- ☐ dilate $\Delta MRC$ using point $C$ as the center
- ☐ dilate $\Delta MRC$ using point $R$ as the center

c. Discuss with your partner what is true about the triangles when all of the transformations have been applied.

**Similarity in terms of non-rigid transformations** is one or more rigid transformations (reflection, rotation, translation) combined with a dilation that leads to two figures being similar.

***Similarity*** is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation, or a sequence of transformations that includes a dilation.

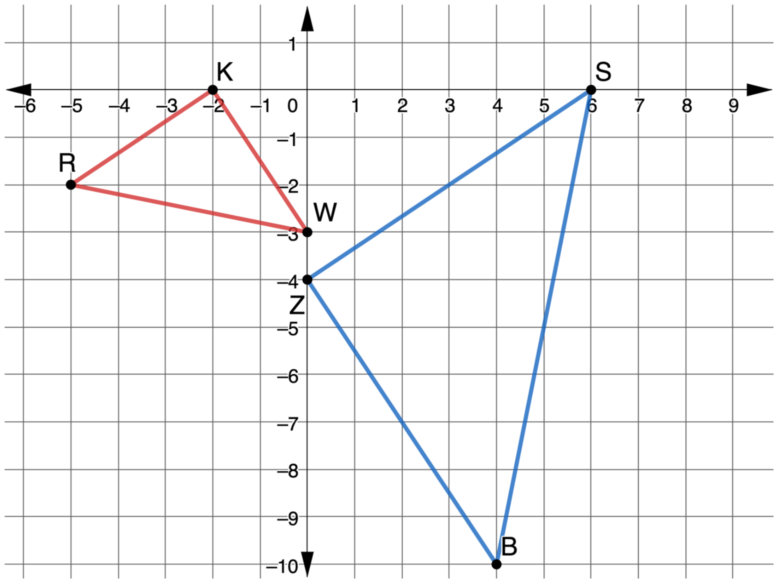
d. Use the definition of similarity in terms of non-rigid motions to justify that $\Delta MRC$ and $\Delta YHC$ are similar.

3. Consider the following statement.

| |
|---|
| If it is possible to find one or more transformations that will transform a figure such that all corresponding points coincide with the other figure, then the figures will be similar. |

Explain what this statement means in your own words.

4. $\Delta RKW$ and $\Delta BZS$ are shown.
Assume: $\frac{RW}{BS} = \frac{WK}{SZ} = \frac{KR}{ZB}$




a. Explain how you can use patty paper to verify $\Delta RKW \sim \Delta BZS$.

b. Write a series of transformations that will map $\Delta RKW$ onto $\Delta BZS$ so that $\Delta RKW \sim \Delta BZS$.

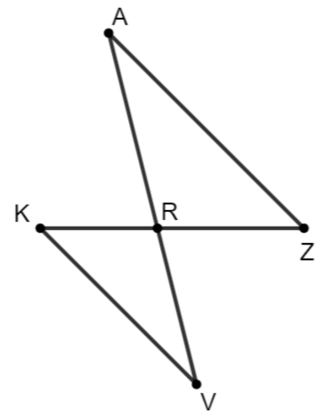
c. If it is given that $\Delta BZS$ is a result of $(x, y) \to (-y, x)$ then $(x, y) \to (2x, 2y)$ to $\Delta RKW$, are the triangles congruent?

d. If it is given that $\Delta BZS$ is a result of $(x, y) \to (-y, x)$ then $(x, y) \to (2x, 2y)$ to $\Delta RKW$, are the triangles similar?

e. Discuss with your partner what criteria would have to be given so that SAS similarity would be shown as sufficient to justify triangles are similar. Summarize your discussion.

5. Katelyn, Diego, and Amelia are writing conclusions about proving similarity in triangles. They used $\Delta KRV$ and $\Delta ZRA$ to write their conclusions. Their conclusions are shown.

Assume: $\angle VKR \cong \angle AZR$ and $\angle VRK \cong \angle ARZ$




| Katelyn's Conclusion | Diego's Conclusion | Amelia's Conclusion |
|---|---|---|
| To prove similarity of two triangles, apply the definition of congruence in terms of rigid motions to show that two angles that are assumed to be congruent are congruent. This is sufficient to prove similarity of two triangles. | Show that two triangles are congruent using rigid motions and dilation. Then, apply the definition of similarity in terms of non-rigid transformations. This is sufficient to prove the triangles are similar. | By applying a series of rigid transformations and a dilation, it could be shown that the triangles are similar by showing that one set of sides are proportional and assumed two angles are congruent. |

a. Whose conclusion do you agree with?

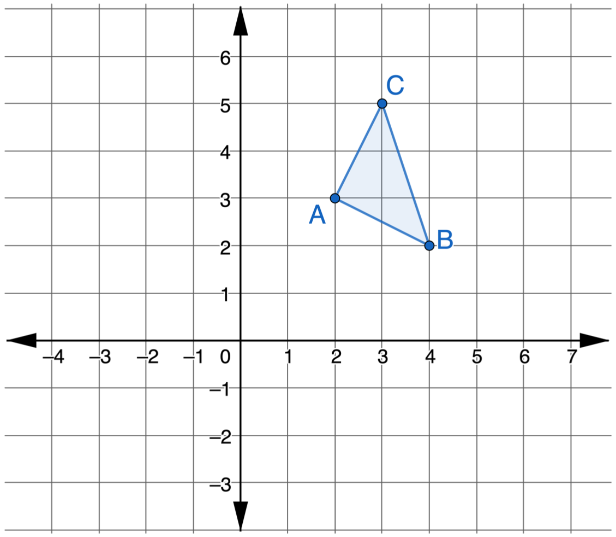
b. Explain why you agree with that student.

c. Pick one student whose conclusion is incorrect. Write a note to them to help them understand their error.

**Means**
**I**
**Start**
**To**
**Acquire**
**Knowledge**
**Experience**
**Skills**

## 6.H3.3 Wrap-Up: Cool Down

1. $\Delta ABC$ is shown.




A sequence of three transformations is shown.

- $(x, y) \rightarrow (x - 4, y - 1)$
- $(x, y) \rightarrow (y, x)$
- $(x, y) \rightarrow (1.5x, 1.5y)$

a. Draw the result of each transformation.

b. Complete the table by noting if the preimage and image are congruent or similar.

| Transformation | Relationship of Preimage to Image |
|---|---|
| $(x, y) \rightarrow (x - 4, y - 1)$ | |
| $(x, y) \rightarrow (y, x)$ | |
| $(x, y) \rightarrow (1.5x, 1.5y)$ | |

# Unit 6, Lesson 10: Similar Figures

### Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

### Learning Target

☐ I can solve mathematical problems using corresponding angles and sides of similar figures.